Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim:- Perform gray level operations images.**	
Open handed problem -7	**Date: 16-08-2026**	**Enrolment No: 92400133079**

Aim :

Perform gray level operation images

For :

Read Digital Image

Convert color image to grayscale image

Image Negation

Image Thresholding

Image Gamma Correction

Gray Scale image

Grayscale images are useful when colors are not important for the task. They make the image simpler and easier for a computer to process, so they are commonly used in image processing and computer vision.

A grayscale image contains only shades of black, gray, and white. Each pixel has one intensity value, usually between 0 and 255 in an 8-bit image:

0 = Black
255 = White

Values between 0 and 255 = Different shades of gray

The value of each pixel tells us how bright or dark that part of the image is. A higher value means a brighter pixel, while a lower value means a darker pixel.

Unlike a color image, a grayscale image does not store separate color information such as red, green, and blue. It mainly represents the brightness of the image, which reduces the amount of data that needs to be processed.

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Applications:

Grayscale images are used in many areas such as medical imaging (X-rays and MRI scans), computer vision, image processing, and document scanning. They are also commonly used in black-and-white photography for artistic purposes.

Grayscale images need less storage space than color images because they contain only one channel for brightness, while a color image usually has three channels (Red, Green, and Blue). This makes grayscale images useful when storage space or processing power is limited.

Code

```
import glob
import os
import cv2
import numpy as np
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt

BASE_DIR = r"D:\SEM 3\python"
IMAGE_DIR = BASE_DIR

files = [
    os.path.join(IMAGE_DIR, f"{i}.jpg")
    for i in range(1, 6)
]
```

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```
if not all(os.path.exists(f) for f in files):
    files = sorted(
        glob.glob(os.path.join(IMAGE_DIR, "*"))
    )[:5]

images = [
    cv2.imread(f, cv2.IMREAD_GRAYSCALE)
    for f in files
]

images = [
    img for img in images
    if img is not None
]

def contrast_stretch(img):
    """Apply contrast stretching to the image."""
    min_val = np.min(img)
    max_val = np.max(img)

    if max_val == min_val:
        return img

    stretched = (
        (img.astype(np.float32) - min_val)
        * 255.0
        / (max_val - min_val)
    )

    return np.uint8(stretched)

def histogram_equalization(img):
    """Apply histogram equalization."""
    return cv2.equalizeHist(img)
```

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```
def gamma_correction(img, gamma=0.4):
    """Apply gamma correction."""
    normalized = img / 255.0

    corrected = np.power(normalized, gamma) * 255.0

    return np.uint8(corrected)

def log_transform(img):
    """Apply logarithmic transformation."""
    normalized = img.astype(np.float32)

    c = 255.0 / np.log(1.0 + np.max(normalized))

    transformed = c * np.log1p(normalized)

    return np.uint8(transformed)

def image_negative(img):
    """Generate the negative of the image."""
    return 255 - img

methods = [
    ("Contrast Stretching", contrast_stretch),
    ("Histogram Equalization", histogram_equalization),
    ("Gamma Transformation (γ = 0.4)", gamma_correction),
    ("Log Transformation", log_transform),
    ("Image Negative", image_negative)
]

enhanced_images = []

for i, (name, func) in enumerate(methods):
```

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```
enhanced = func(images[i])

    enhanced_images.append(enhanced)

    output_path = os.path.join(
        BASE_DIR,
        f"enhanced_image_{i + 1}.png"
    )

    cv2.imwrite(output_path, enhanced)

fig, axes = plt.subplots(
    5,
    2,
    figsize=(10, 18)
)

for i in range(5):

    name, _ = methods[i]

    axes[i, 0].imshow(
        images[i],
        cmap="gray"
    )

    axes[i, 0].set_title(
        f"Image {i + 1} - Original",
        fontsize=11,
        fontweight="bold"
    )

    axes[i, 0].axis("off")
```

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```
axes[i, 1].imshow(
        enhanced_images[i],
        cmap="gray"
    )

    axes[i, 1].set_title(
        f"Image {i + 1} - {name}",
        fontsize=11,
        fontweight="bold"
    )

    axes[i, 1].axis("off")

plt.tight_layout()

plt.savefig(
    os.path.join(BASE_DIR, "comparison_results.png"),
    dpi=300,
    bbox_inches="tight"
)

plt.close()

fig_hist, axes_hist = plt.subplots(
    5,
    2,
    figsize=(12, 16)
)

for i in range(5):

    name, _ = methods[i]

    axes_hist[i, 0].hist(
```

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```
 images[i].ravel(),
        bins=256,
        range=[0, 256],
        color="gray",
        alpha=0.8
    )

    axes_hist[i, 0].set_title(
        f"Image {i + 1} - Original Histogram",
        fontsize=10
    )

    axes_hist[i, 0].set_xlim([0, 256])

    axes_hist[i, 1].hist(
        enhanced_images[i].ravel(),
        bins=256,
        range=[0, 256],
        color="indigo",
        alpha=0.8
    )

    axes_hist[i, 1].set_title(
        f"Image {i + 1} - Enhanced Histogram ({name})",
        fontsize=10
    )

    axes_hist[i, 1].set_xlim([0, 256])

plt.tight_layout()

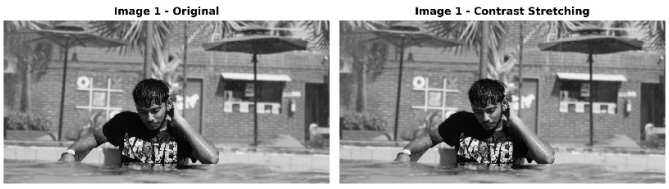
plt.savefig(
    os.path.join(BASE_DIR, "histogram_results.png"),
    dpi=300,
```

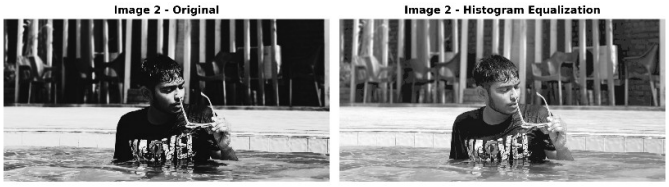
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```
 bbox_inches="tight"
)

plt.close()
```

output
comparison image











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Enhanced image 1



Enhanced 2



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Enhanced image 3



Enhanced image 4



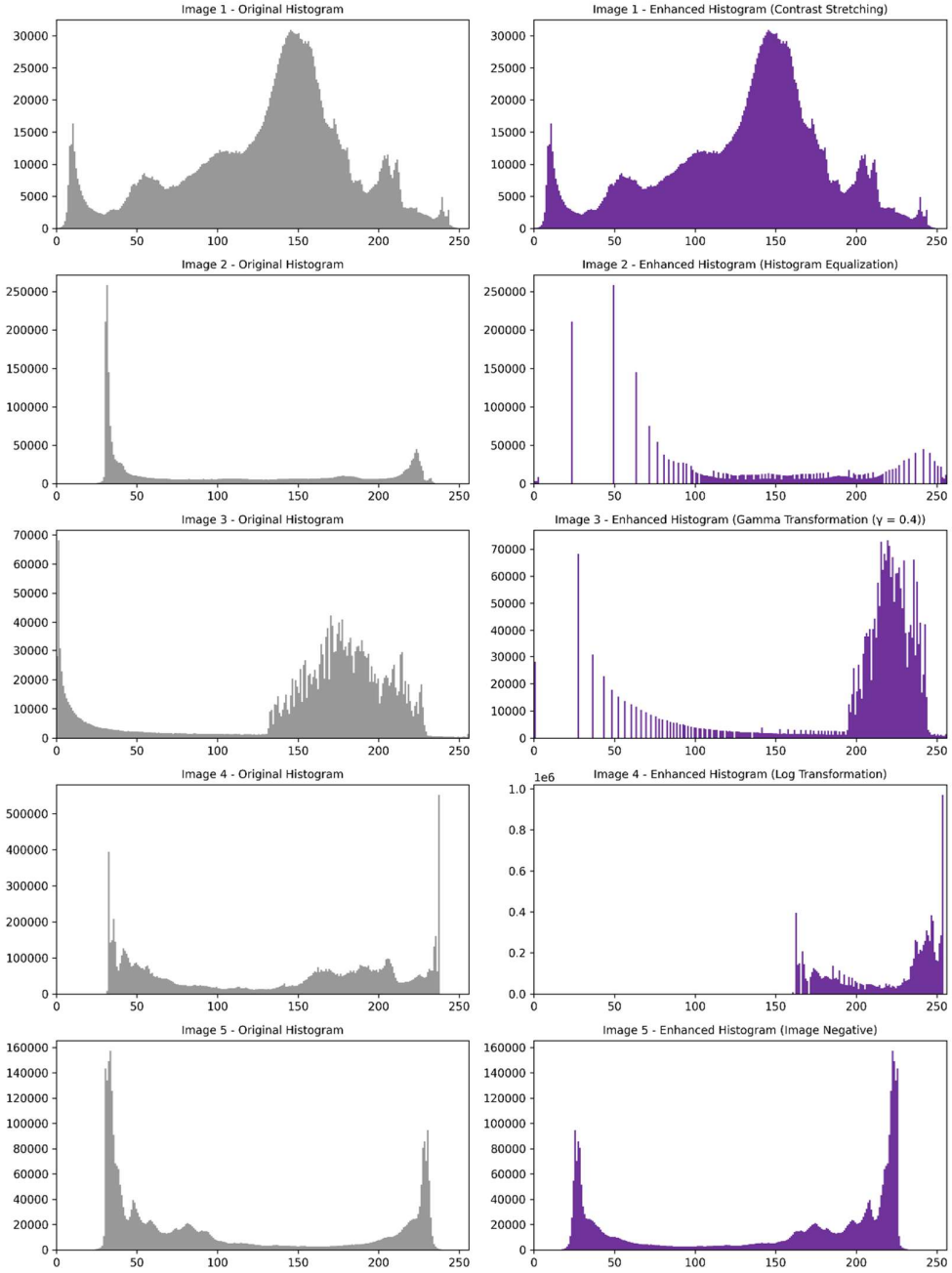
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Enhanced image 5



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Histogram results



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Conclusion

Different image enhancement techniques were successfully implemented using Python and OpenCV.

Contrast stretching improved the intensity range and enhanced image details.

Histogram equalization improved the overall contrast of the images.

Gamma transformation adjusted the brightness of the images effectively.

Log transformation enhanced details, especially in darker regions.

Image negative produced the inverse intensity of the original image.

Original and enhanced images were displayed side by side for comparison.

Histograms were plotted to analyze the distribution of pixel intensities.

Each enhancement technique produced different visual results.

The experiment showed that image enhancement improves image quality and visibility.

OpenCV provided efficient functions for image processing operations.

Matplotlib was used to visualize images and their histograms.

The experiment provided practical understanding of spatial-domain image enhancement techniques.

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